

Cambridge (CIE) IGCSE Chemistry
Paper 3: Theory (Set A)
Core

Saturday 26 April 2025

Morning (Time: 1 hour 15 minutes)

Total marks

/ 80

Instructions

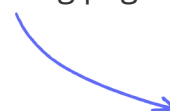
- Try to complete this mock exam paper in one sitting, under exam conditions. Use all the time available and check your answers to each question at the end before submitting.
- Remember this is PRACTICE. Mistakes are fine and will help you improve in time for the real exam - just do your best.
- You may use a calculator.
- You should show all your working and use appropriate units.

Materials

- A copy of [the Periodic Table of elements](#) is required for this paper.

Scan here to mark your mock exam

or visit the mock exams landing page for this course



1 (a) The table gives the composition of three particles.

particle	number of protons	number of electrons	number of neutrons
A	15	15	16
B	15	15	16
C	15	15	17

What is the evidence in the table for each of the following?

i) Particle **A** is an atom.

[1]

ii) **A**, **B** and **C** are all particles of the same element.

[1]

iii) Particles **A** and **C** are isotopes of the same element.

[2]

(4 marks)

(b) i) What is the electronic structure of particle **A**?

[1]

ii) Is element **A**, a metal or a non-metal? Give a reason for your choice.

[1]

(2 marks)

2 (a) This question is about ammonia and ammonium salts.

Ammonia is a gas at room temperature. What is the test for ammonia gas? Describe the positive result of this test.

test

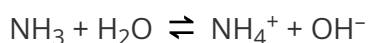
result

.....

.....

(2 marks)

(b) Ammonia reacts with water to form ions.



Describe what is seen when aqueous ammonia is added to aqueous copper(II) sulfate, until no further change is seen.

.....

.....

.....

(3 marks)

(c) Barium sulfate is an insoluble salt.

Barium sulfate can be made from aqueous ammonium sulfate using a precipitation reaction.

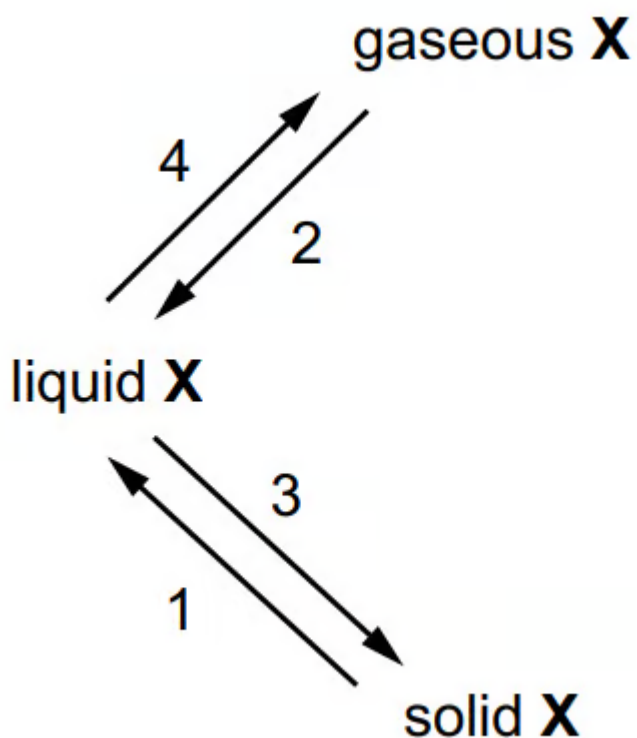
Name a solution that can be added to aqueous ammonium sulfate to produce a precipitate of barium sulfate.

.....

(1 mark)

3 (a) Element X can undergo the following physical changes.

Figure 1.1



Name each of the numbered physical changes shown in **Figure 1**.

(4 marks)

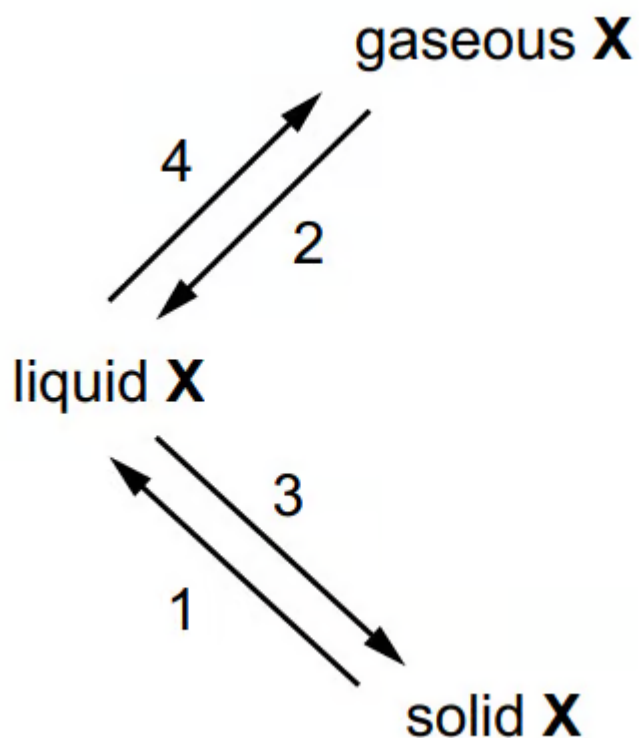
(b) One difference between boiling and evaporation is the rate at which the processes occur.

State one other difference between boiling and evaporation.

(1 mark)

(c) Element X can undergo the following physical changes.

Figure 1.1



Describe the separation, arrangement and motion of particles of element X in the solid state.

Separation:

Arrangement:

Motion:

.....

.....

.....

(3 marks)

4 (a) The table below gives the number of protons, neutrons and electrons in atoms or ions.

Complete the table. The first line is given as an example.

You will need to use the Periodic Table.

particle	number of protons	number of electrons	number of neutrons	symbol or formula
A	4	4	5	${}^9_4\text{Be}$
B	19	18	20	
C	30	30	35	
D	8	10	8	
E	31	31	39	

.....

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.....

(6 marks)

(b) Using the data in the table, explain how you can determine whether a particle is an atom, a negative ion or a positive ion.

.....

.....

.....

(3 marks)

- 5 (a)** Manganese is a transition element. It has more than one valency and the metal and its compounds are catalysts.

Predict three other properties of manganese that are typical of transition elements.

(3 marks)

- (b)** It has several oxides, three of which are shown below. Manganese(II) oxide, which is basic. Manganese(III) oxide, which is amphoteric. Manganese(IV) oxide, which is acidic.

i) Complete the word equation.

manganese(II) oxide + hydrochloric acid → +

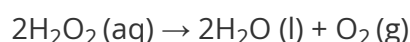
[2]

ii) Which, if any, of these oxides will react with sodium hydroxide?

[1]

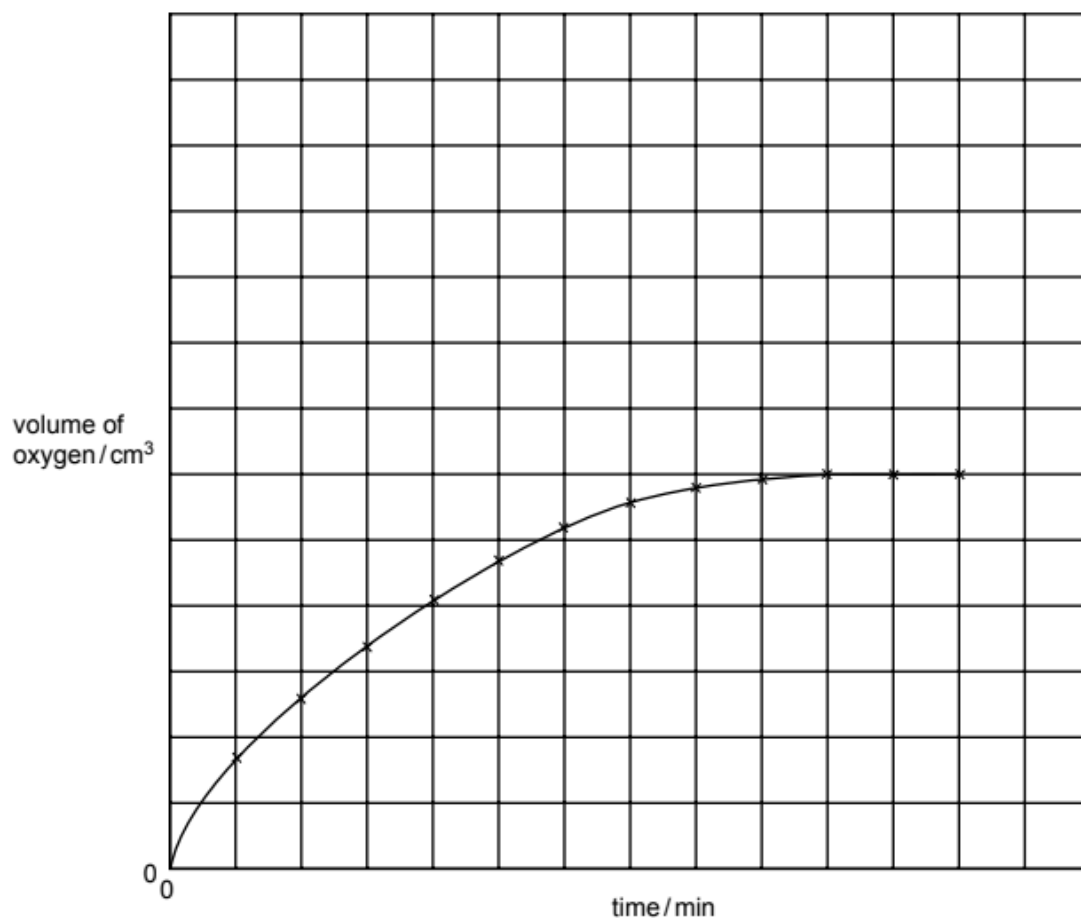
(3 marks)

- (c)** Aqueous hydrogen peroxide decomposes to form water and oxygen.



This reaction is catalysed by manganese(IV) oxide. The following experiments were carried out to investigate the rate of this reaction. A 0.1 g sample of manganese(IV) oxide was added to 20 cm³ of 0.2 mol / dm³ hydrogen peroxide solution. The volume of oxygen

produced was measured every minute. The results of this experiment are shown on the graph.



i) How does the rate of reaction vary with time? Explain why the rate varies.

[3]

ii) The following experiment was carried out at the same temperature. 0.1 g of manganese(IV) oxide and 20 cm³ of 0.4 mol / dm³ hydrogen peroxide. Sketch the curve for this experiment on the same grid.

[2]

iii) How would the shape of the graph differ if only half the mass of catalyst had been used in these experiments?

[2]

(7 marks)

6 (a) The properties of five alkenes at room temperature are shown in **Table 1.1**

Table 1.1

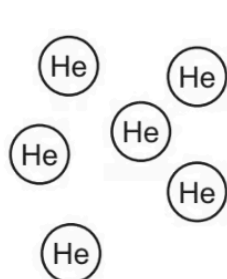
alkene	number of carbon atoms in a molecule	state at room temperature	density in g / cm ³	boiling point / °C
ethene	2	gas	0.0012	-104
propene	3	gas	0.0018	-47
butene	4	gas	0.0024	
pentene	5	liquid	0.64	30
hexene	6	liquid	0.67	63

Answer this question using only the information in the table.

Suggest why the densities of the first three alkenes are much lower than the density of pentene and hexene.

(1 mark)

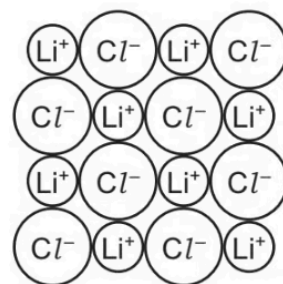
(b) The diagrams show part of the structures of five substances, A, B, C, D and E.



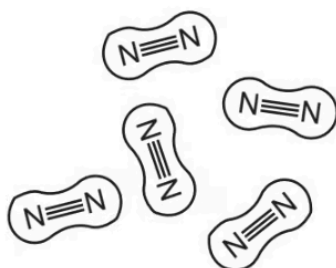
A



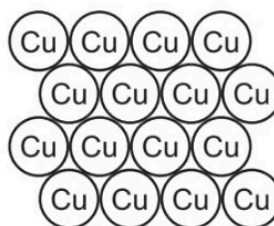
B



C



D



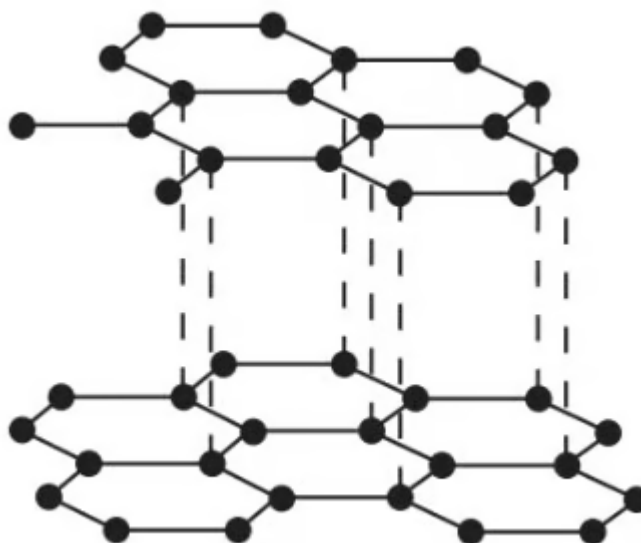
E

Answer the following question about these structures.

Which one of these structures, A, B, C, D or E, is used in cutting tools?

(1 mark)

(c) Part of the structure of graphite is shown.



Use the information from the diagram to explain why graphite is used as a lubricant.

.....
(1 mark)

7 (a) Aqueous sodium hydroxide is a base.

Complete this sentence about the different types of bases. Bases are metal hydroxides or metal _____

.....
(1 mark)

(b) Describe the reaction of aqueous sodium hydroxide with:

i) A named acid

ii) An ammonium salt

.....
.....
.....
.....
(4 marks)

(c) Ammonia is a soluble base.

Draw a circle around the pH value of aqueous ammonia.

pH 1 pH 5 pH 7 pH 10

.....
(1 mark)

8 (a) Three of the halogens in Group VII are listed below.

chlorine
bromine
iodine

i) How does their colour change down the Group?

[1]

ii) How do their melting points and boiling points change down the Group?

[1]

iii) Predict the colour and physical state (solid, liquid or gas) of astatine, At. colour ...
physical state ...

[2]

.....

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.....

.....

(4 marks)

(b) A radioactive isotope of iodine, $^{131}_{53}\text{I}$, is used to treat cancer.

i) Define the term *isotope*.

[2]

ii) How many protons, electrons and neutrons are there in one atom of $^{131}_{53}\text{I}$

number of protons number of electrons number of neutrons

[2]

iii) When this isotope, $^{131}_{53}\text{I}$, emits radiation, a different element with a proton number of 54 is formed. What is the name of this element?

[1]

(5 marks)

9 (a) This question is concerned with the following oxides.

sulfur dioxide
carbon monoxide
lithium oxide
aluminium oxide
nitrogen dioxide
strontium oxide

Two of the oxides are responsible for acid rain.

Identify the two oxides and explain their presence in the atmosphere.

(5 marks)

(b) Lithium oxide is an ionic compound.

i) Identify another ionic oxide in the list in part a).

[1]

ii) Draw a diagram which shows the formula of lithium oxide, the charges on the ions and the arrangement of the valency electrons around the negative ion.

Use x to represent an electron from an atom of oxygen. Use o to represent an electron from an atom of lithium.

[2]

(3 marks)

10 (a) Two homologous series of hydrocarbons are the alkanes and the alkenes.

One general characteristic of a homologous series is that the physical properties vary in a predictable way.

i) State **three** other general characteristics of a homologous series.

[3]

ii) How can the molecular formula of a hydrocarbon show whether it is an alkane or an alkene?

[2]

iii) How do alkanes and alkenes differ in their molecular structures?

[2]

(7 marks)

(b) Cracking is the thermal decomposition of alkanes into smaller hydrocarbons and possibly hydrogen.

i) State **two** conditions required for the cracking of an alkane.

[2]

ii) One type of cracking produces an alkane and an alkene. Complete an equation for the cracking of heptane into an alkane and an alkene.



[1]

iii) Complete an equation for the cracking of heptane into hydrogen and two other products.



[1]

iv) Suggest **one** reason why cracking is important.

[1]

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.....

(5 marks)